Lesson 4. Systems of Differential Equations I

ELTE Eötvös Loránd University

Faculty of Informatics, Department of Numerical Analysis

2024

Table of Contents

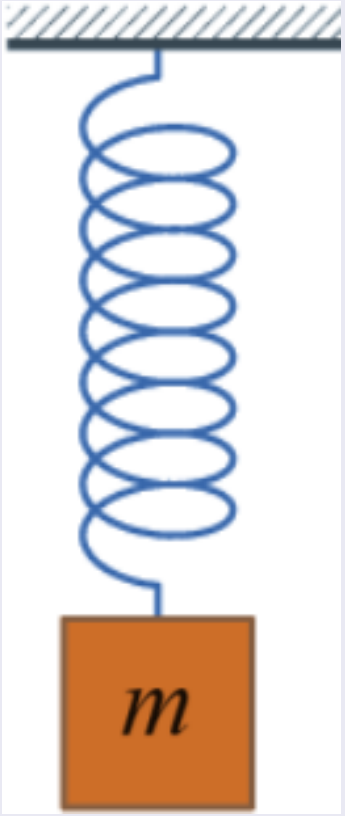
1. 4.1. Basic theory of systems of ODEs
2. 4.2. Equilibria Points and Basic Concepts of Stability Theory
3. 4.3. Equilibrium Points of Linear Autonomous Systems and its Stability

Example 1: Damped Harmonic Oscillator Equation

Consider the equation for a damped harmonic oscillator:

$$m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = 0$$

- m is the mass of the oscillator.
- $x(t)$ is the displacement of the oscillator as a function of time t.
- b is the damping coefficient, representing the damping force proportional to velocity.
- k is the spring constant, representing the restoring force proportional to displacement.



Example 1: Damped Harmonic Oscillator Equation

$$m\frac{d^2x(t)}{dt^2} + b\frac{dx(t)}{dt} + kx(t) = 0$$

This equation can be rewritten as a so-called system of first order ODEs. This is a common technique in differential equations that can be achieved using substitution. We introduce new variables:

$$x_1 = x \text{ and } x_2 = \frac{dx}{dt}$$

Then, the original second-order ODE can be transformed into the following system of first-order ODEs:

$$\frac{dx_1}{dt} = x_2$$

$$\frac{dx_2}{dt} = -\frac{b}{m}x_2 - \frac{k}{m}x_1$$

Here, x_1 represents the position x, and x_2 represents the velocity $\frac{dx}{dt}$. This system of first-order ODEs provides an equivalent description of the dynamics of the damped harmonic oscillator.

Example 1: Damped Harmonic Oscillator Equation

$$m\frac{d^2x(t)}{dt^2} + b\frac{dx(t)}{dt} + kx(t) = 0$$

This equation can be rewritten as a so-called system of first order ODEs. This is a common technique in differential equations that can be achieved using substitution. We introduce new variables:

$$x_1 = x \text{ and } x_2 = \frac{dx}{dt}$$

Then, the original second-order ODE can be transformed into the following system of first-order ODEs:

$$\frac{dx_1}{dt} = x_2$$

$$\frac{dx_2}{dt} = -\frac{b}{m}x_2 - \frac{k}{m}x_1$$

Here, x_1 represents the position x, and x_2 represents the velocity $\frac{dx}{dt}$. This system of first-order ODEs provides an equivalent description of the dynamics of the damped harmonic oscillator.

Example 1: Damped Harmonic Oscillator Equation

$$m\frac{d^2x(t)}{dt^2} + b\frac{dx(t)}{dt} + kx(t) = 0$$

This equation can be rewritten as a so-called system of first order ODEs. This is a common technique in differential equations that can be achieved using substitution. We introduce new variables:

$$x_1 = x \text{ and } x_2 = \frac{dx}{dt}$$

Then, the original second-order ODE can be transformed into the following system of first-order ODEs:

$$\frac{dx_1}{dt} = x_2$$

$$\frac{dx_2}{dt} = -\frac{b}{m}x_2 - \frac{k}{m}x_1$$

Here, x_1 represents the position x, and x_2 represents the velocity $\frac{dx}{dt}$. This system of first-order ODEs provides an equivalent description of the dynamics of the damped harmonic oscillator.

Example 1: Damped Harmonic Oscillator Equation

$$m\frac{d^2x(t)}{dt^2} + b\frac{dx(t)}{dt} + kx(t) = 0$$

This equation can be rewritten as a so-called system of first order ODEs. This is a common technique in differential equations that can be achieved using substitution. We introduce new variables:

$$x_1 = x \text{ and } x_2 = \frac{dx}{dt}$$

Then, the original second-order ODE can be transformed into the following system of first-order ODEs:

$$\frac{dx_1}{dt} = x_2$$

$$\frac{dx_2}{dt} = -\frac{b}{m}x_2 - \frac{k}{m}x_1$$

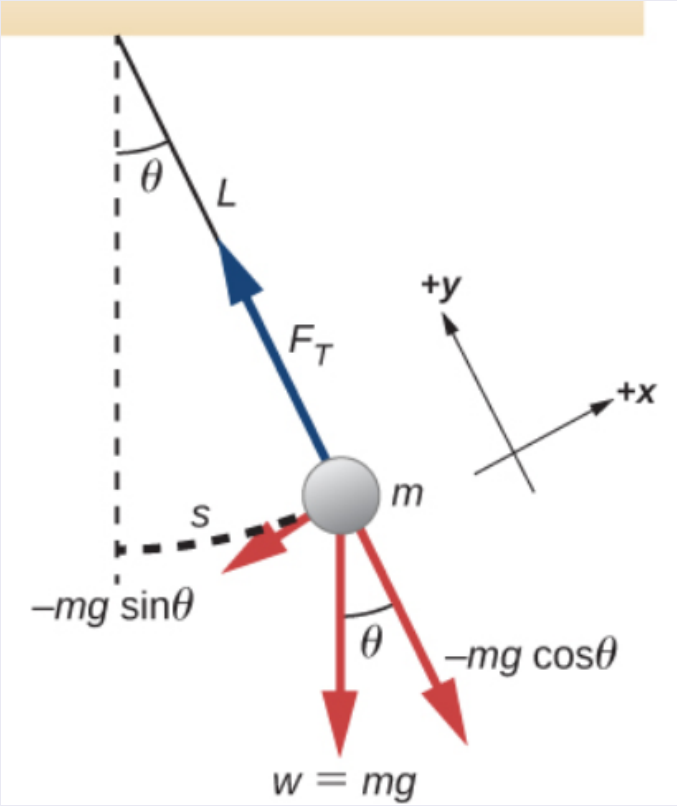
Here, x_1 represents the position x, and x_2 represents the velocity $\frac{dx}{dt}$. This system of first-order ODEs provides an equivalent description of the dynamics of the damped harmonic oscillator.

Example 2: Simple Pendulum Equation

The swinging of a pendulum is governed by the equation:

$$\frac{d^2x}{dt^2} + \frac{g}{L}\sin(x) = 0$$

- $x(t)$: Angular displacement of the pendulum as a function of time t.
- g: Acceleration due to gravity (approximately $9.81\,\mathrm{m/s^2}$).
- L: Length of the pendulum from the pivot to the center of mass.
- $\sin(x)$: Sine of the angular displacement x, representing the restoring force.



Example 2: Simple Pendulum Equation

$$\frac{d^2x}{dt^2} + \frac{g}{L}\sin(x) = 0$$

This equation can also be rewritten as a system of first-order ODEs using substitution.

We introduce new variables:

$$x_1 = x \quad \text{and} \quad x_2 = \frac{dx}{dt}$$

Then, the original second-order ODE can be transformed into the following system of first-order ODEs:

$$\frac{dx_1}{dt} = x_2$$

$$\frac{dx_2}{dt} = -\frac{g}{L}\sin(x_1)$$

Here, x_1 represents the angular displacement x of the pendulum, and x_2 represents the angular velocity $\frac{dx}{dt}$. This system of first-order ODEs provides an equivalent description of the dynamics of the simple pendulum.

Example 2: Simple Pendulum Equation

$$\frac{d^2x}{dt^2} + \frac{g}{L}\sin(x) = 0$$

This equation can also be rewritten as a system of first-order ODEs using substitution.
We introduce new variables:

$$x_1 = x \quad \text{and} \quad x_2 = \frac{dx}{dt}$$

Then, the original second-order ODE can be transformed into the following system of first-order ODEs:

$$\frac{dx_1}{dt} = x_2$$

$$\frac{dx_2}{dt} = -\frac{g}{L}\sin(x_1)$$

Here, x_1 represents the angular displacement x of the pendulum, and x_2 represents the angular velocity $\frac{dx}{dt}$. This system of first-order ODEs provides an equivalent description of the dynamics of the simple pendulum.

Example 2: Simple Pendulum Equation

$$\frac{d^2x}{dt^2} + \frac{g}{L}\sin(x) = 0$$

This equation can also be rewritten as a system of first-order ODEs using substitution.
We introduce new variables:

$$x_1 = x \quad \text{and} \quad x_2 = \frac{dx}{dt}$$

Then, the original second-order ODE can be transformed into the following system of first-order ODEs:

$$\frac{dx_1}{dt} = x_2$$

$$\frac{dx_2}{dt} = -\frac{g}{L}\sin(x_1)$$

Here, x_1 represents the angular displacement x of the pendulum, and x_2 represents the angular velocity $\frac{dx}{dt}$. This system of first-order ODEs provides an equivalent description of the dynamics of the simple pendulum.

Example 2: Simple Pendulum Equation

$$\frac{d^2x}{dt^2} + \frac{g}{L}\sin(x) = 0$$

This equation can also be rewritten as a system of first-order ODEs using substitution.
We introduce new variables:

$$x_1 = x \quad \text{and} \quad x_2 = \frac{dx}{dt}$$

Then, the original second-order ODE can be transformed into the following system of first-order ODEs:

$$\frac{dx_1}{dt} = x_2$$

$$\frac{dx_2}{dt} = -\frac{g}{L}\sin(x_1)$$

Here, x_1 represents the angular displacement x of the pendulum, and x_2 represents the angular velocity $\frac{dx}{dt}$. This system of first-order ODEs provides an equivalent description of the dynamics of the simple pendulum.

Basic theory of systems of DEs

A system of differential equations is a finite set of differential equations.

Classification

- systems of ODEs/PDEs;
- Order/degree;
- Linearity

Definition of a System of First-Order ODEs

A system of first-order ordinary differential equations (ODEs) can be expressed in the standard form as follows:

$$\begin{cases} x_1' = f_1(t, x_1, x_2, \ldots, x_n) \\ x_2' = f_2(t, x_1, x_2, \ldots, x_n) \\ \vdots \\ x_n' = f_n(t, x_1, x_2, \ldots, x_n) \end{cases}$$

Basic theory of systems of DEs

A system of differential equations is a finite set of differential equations.

Classification

- systems of ODEs/PDEs;
- Order/degree;
- Linearity

Definition of a System of First-Order ODEs

A system of first-order ordinary differential equations (ODEs) can be expressed in the standard form as follows:

$$\begin{cases} x_1' = f_1(t, x_1, x_2, \ldots, x_n) \\ x_2' = f_2(t, x_1, x_2, \ldots, x_n) \\ \vdots \\ x_n' = f_n(t, x_1, x_2, \ldots, x_n) \end{cases}$$

Basic theory of systems of DEs

A system of differential equations is a finite set of differential equations.

Classification

- systems of ODEs/PDEs;
- Order/degree;
- Linearity

Definition of a System of First-Order ODEs

A system of first-order ordinary differential equations (ODEs) can be expressed in the standard form as follows:

$$\begin{cases} x_1' = f_1(t, x_1, x_2, \ldots, x_n) \\ x_2' = f_2(t, x_1, x_2, \ldots, x_n) \\ \vdots \\ x_n' = f_n(t, x_1, x_2, \ldots, x_n) \end{cases}$$

Basic theory of systems of DEs

A system of differential equations is a finite set of differential equations.

Classification

- systems of ODEs/PDEs;
- Order/degree;
- Linearity

Definition of a System of First-Order ODEs

A system of first-order ordinary differential equations (ODEs) can be expressed in the standard form as follows:

$$\begin{cases} x_1' = f_1(t, x_1, x_2, \ldots, x_n) \\ x_2' = f_2(t, x_1, x_2, \ldots, x_n) \\ \vdots \\ x_n' = f_n(t, x_1, x_2, \ldots, x_n) \end{cases}$$

Basic theory of systems of DEs

A system of differential equations is a finite set of differential equations.

Classification

- systems of ODEs/PDEs;
- Order/degree;
- Linearity

Definition of a System of First-Order ODEs

A system of first-order ordinary differential equations (ODEs) can be expressed in the standard form as follows:

$$\begin{cases} x_1' = f_1(t, x_1, x_2, \ldots, x_n) \\ x_2' = f_2(t, x_1, x_2, \ldots, x_n) \\ \vdots \\ x_n' = f_n(t, x_1, x_2, \ldots, x_n) \end{cases}$$

Basic theory of systems of DEs

A system of ODEs is said to be linear if each function f_i is linear in all the dependent variables $x_1, x_2, \ldots, x_n$. A system is non-linear if any of the functions f_i involve nonlinear terms in the dependent variables.

Example of a Linear System:

$$\begin{cases} x_1' = x_2 \\ x_2' = -\frac{k}{m}x_1 - \frac{b}{m}x_2 \end{cases}$$

Example of a Non-Linear System:

$$\begin{cases} x_1' = x_2 \\ x_2' = -\frac{g}{L}\sin(x_1) \end{cases}$$

Solution to a System of First-Order ODEs

A system of functions

$$\begin{cases} x_1(t) = \Phi_1(t) \\ x_2(t) = \Phi_2(t) \\ \vdots \\ x_n(t) = \Phi_n(t) \end{cases}$$

is a solution to the system of differential equations if each function $\Phi_j(t)$ for $1 \leq j \leq n$ is continuously differentiable on I and satisfies the corresponding differential equation on I.

Basic theory of systems of DEs

A system of ODEs is said to be linear if each function f_i is linear in all the dependent variables $x_1, x_2, \ldots, x_n$. A system is non-linear if any of the functions f_i involve nonlinear terms in the dependent variables.

Example of a Linear System:

$$\begin{cases} x_1' = x_2 \\ x_2' = -\frac{k}{m}x_1 - \frac{b}{m}x_2 \end{cases}$$

Example of a Non-Linear System:

$$\begin{cases} x_1' = x_2 \\ x_2' = -\frac{g}{L}\sin(x_1) \end{cases}$$

Solution to a System of First-Order ODEs

A system of functions

$$\begin{cases} x_1(t) = \Phi_1(t) \\ x_2(t) = \Phi_2(t) \\ \vdots \\ x_n(t) = \Phi_n(t) \end{cases}$$

is a solution to the system of differential equations if each function $\Phi_j(t)$ for $1 \leq j \leq n$ is continuously differentiable on I and satisfies the corresponding differential equation on I.

Basic theory of systems of DEs

A system of ODEs is said to be linear if each function f_i is linear in all the dependent variables $x_1, x_2, \ldots, x_n$. A system is non-linear if any of the functions f_i involve nonlinear terms in the dependent variables.

Example of a Linear System:

$$\begin{cases} x_1' = x_2 \\ x_2' = -\frac{k}{m}x_1 - \frac{b}{m}x_2 \end{cases}$$

Example of a Non-Linear System:

$$\begin{cases} x_1' = x_2 \\ x_2' = -\frac{g}{L}\sin(x_1) \end{cases}$$

Solution to a System of First-Order ODEs

A system of functions

$$\begin{cases} x_1(t) = \Phi_1(t) \\ x_2(t) = \Phi_2(t) \\ \vdots \\ x_n(t) = \Phi_n(t) \end{cases}$$

is a solution to the system of differential equations if each function $\Phi_j(t)$ for $1 \leq j \leq n$ is continuously differentiable on I and satisfies the corresponding differential equation on I.

Basic theory of systems of DEs

A system of ODEs is said to be linear if each function f_i is linear in all the dependent variables $x_1, x_2, \ldots, x_n$. A system is non-linear if any of the functions f_i involve nonlinear terms in the dependent variables.

Example of a Linear System:

$$\begin{cases} x_1' = x_2 \\ x_2' = -\frac{k}{m}x_1 - \frac{b}{m}x_2 \end{cases}$$

Example of a Non-Linear System:

$$\begin{cases} x_1' = x_2 \\ x_2' = -\frac{g}{L}\sin(x_1) \end{cases}$$

Solution to a System of First-Order ODEs

A system of functions

$$\begin{cases} x_1(t) = \Phi_1(t) \\ x_2(t) = \Phi_2(t) \\ \vdots \\ x_n(t) = \Phi_n(t) \end{cases}$$

is a solution to the system of differential equations if each function $\Phi_j(t)$ for $1 \leq j \leq n$ is continuously differentiable on I and satisfies the corresponding differential equation on I.

Basic theory of systems of DEs

Initial Value Problem (IVP) for a System of First-Order ODEs

An initial value problem (IVP) for a system of first-order ordinary differential equations (ODEs) consists of the System of Differential Equations and Initial Conditions:

$$\begin{cases} x_1(t_0) = k_1 \\ x_2(t_0) = k_2 \\ \vdots \\ x_n(t_0) = k_n \end{cases}$$

where:

- t_0 is the initial time.
- $k_1, k_2, \ldots, k_n$ are the initial values of the functions $x_1, x_2, \ldots, x_n$ at $t = t_0$.

Solution to the IVP

A solution to the IVP is a set of functions $\{x_1(t), x_2(t), \ldots, x_n(t)\}$ that satisfies the system of differential equations and the initial conditions for t in some interval containing t_0.

Basic theory of systems of DEs

Initial Value Problem (IVP) for a System of First-Order ODEs

An initial value problem (IVP) for a system of first-order ordinary differential equations (ODEs) consists of the System of Differential Equations and Initial Conditions:

$$\begin{cases} x_1(t_0) = k_1 \\ x_2(t_0) = k_2 \\ \vdots \\ x_n(t_0) = k_n \end{cases}$$

where:

- t_0 is the initial time.
- $k_1, k_2, \ldots, k_n$ are the initial values of the functions $x_1, x_2, \ldots, x_n$ at $t = t_0$.

Solution to the IVP

A solution to the IVP is a set of functions $\{x_1(t), x_2(t), \ldots, x_n(t)\}$ that satisfies the system of differential equations and the initial conditions for t in some interval containing t_0.

Basic theory of systems of DEs

Theorem: Existence and Uniqueness Theorem

Let

$$f_i, \quad 1 \leq i \leq n \text{ and } \frac{\partial f_i}{\partial x_j}, \quad 1 \leq i \leq n, 1 \leq j \leq n$$

be continuous in some domain R containing the point $(t_0, k_1, \ldots, k_n)$.
Then the initial value problem has a unique solution on some interval $(t_0 - \alpha, t_0 + \alpha)$.

Theorem: Existence and Uniqueness Theorem for Linear Systems

Consider the linear system of differential equations

$$\mathbf{x}'(t) = A(t)\mathbf{x}(t) + \mathbf{b}(t)$$

If the functions $a_{ij}(t)$ and $b_i(t)$ are continuous on an interval $(t_0 - \alpha, t_0 + \alpha)$, then the initial value problem

$$\mathbf{x}(t_0) = \mathbf{x}_0$$

has a unique solution on this interval.

Basic theory of systems of DEs

Theorem: Existence and Uniqueness Theorem

Let

$$f_i, \quad 1 \leq i \leq n \text{ and } \frac{\partial f_i}{\partial x_j}, \quad 1 \leq i \leq n, 1 \leq j \leq n$$

be continuous in some domain R containing the point $(t_0, k_1, \ldots, k_n)$.
Then the initial value problem has a unique solution on some interval $(t_0 - \alpha, t_0 + \alpha)$.

Theorem: Existence and Uniqueness Theorem for Linear Systems

Consider the linear system of differential equations

$$\mathbf{x}'(t) = A(t)\mathbf{x}(t) + \mathbf{b}(t)$$

If the functions $a_{ij}(t)$ and $b_i(t)$ are continuous on an interval $(t_0 - \alpha, t_0 + \alpha)$, then the initial value problem

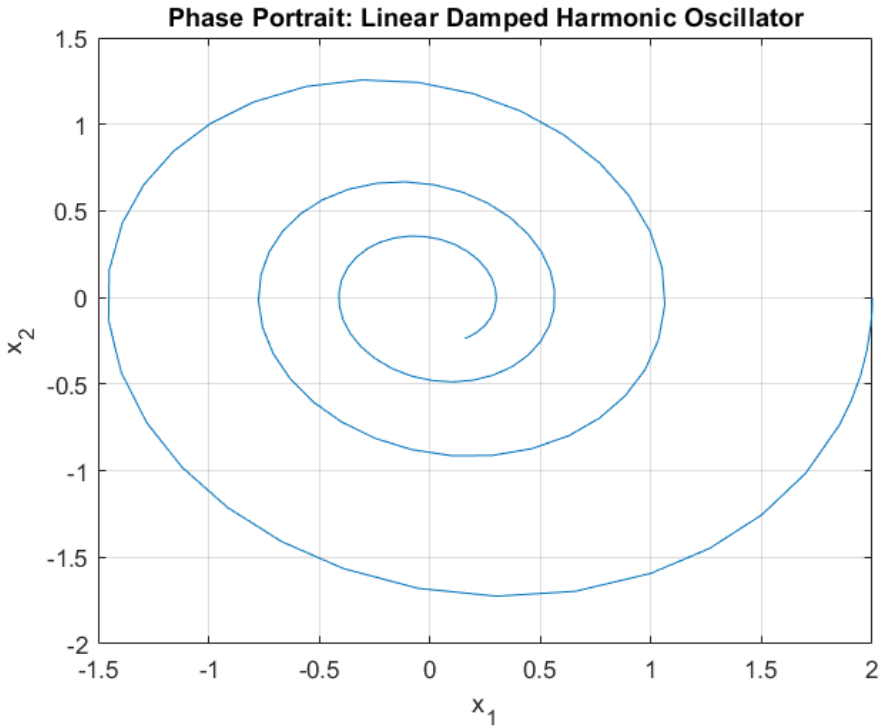
$$\mathbf{x}(t_0) = \mathbf{x}_0$$

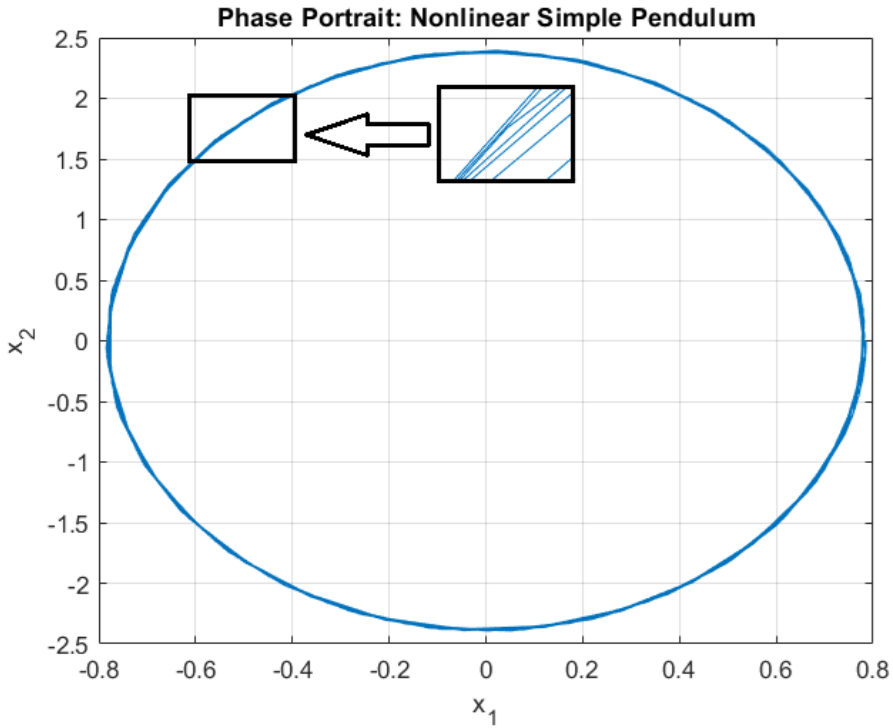
has a unique solution on this interval.

Basic theory of systems of DEs

How to visualize solution curves in the solution space?

The solution space is called a phase space (phase portrait), and the solution curve $\mathbf{x} = x_j(t)|_{j=1}^{n}$ is called the trajectory or the orbit in the phase space $\mathbb{R}^n$.





Qualitative analysis of systems of DEs

Our goal is to run this construction in reverse:

given the system, we want to draw the trajectories, and thereby extract information about the solutions.

This approach is called qualitative analysis of differential equations, and it is a powerful method for understanding the behavior of systems without needing to solve the equations explicitly.

Qualitative analysis of systems of DEs

- Finding Equilibrium Points;
- Determining Stability;
- Global Behavior Analysis;
- Identifying the Phase Space and Sketching the Trajectories

Qualitative analysis of systems of DEs

Our goal is to run this construction in reverse:

given the system, we want to draw the trajectories, and thereby extract information about the solutions.

This approach is called qualitative analysis of differential equations, and it is a powerful method for understanding the behavior of systems without needing to solve the equations explicitly.

Qualitative analysis of systems of DEs

- Finding Equilibrium Points;
- Determining Stability;
- Global Behavior Analysis;
- Identifying the Phase Space and Sketching the Trajectories

Qualitative analysis of systems of DEs

Our goal is to run this construction in reverse:

given the system, we want to draw the trajectories, and thereby extract information about the solutions.

This approach is called qualitative analysis of differential equations, and it is a powerful method for understanding the behavior of systems without needing to solve the equations explicitly.

Qualitative analysis of systems of DEs

- Finding Equilibrium Points;
- Determining Stability;
- Global Behavior Analysis;
- Identifying the Phase Space and Sketching the Trajectories

Table of Contents

1. 4.1. Basic theory of systems of ODEs
2. 4.2. Equilibria Points and Basic Concepts of Stability Theory
3. 4.3. Equilibrium Points of Linear Autonomous Systems and its Stability

Equilibria Points

Equilibria points (or equilibrium points, critical points, stationary points) of a system of differential equations are points in the phase space where the system does not change over time.

Mathematically, for a system of differential equations:

$$\begin{cases} x_1' = f_1(x_1, x_2, \ldots, x_n) \\ x_2' = f_2(x_1, x_2, \ldots, x_n) \\ \vdots \\ x_n' = f_n(x_1, x_2, \ldots, x_n) \end{cases}$$

an equilibrium point

$$\mathbf{x}^* = (x_1^*, x_2^*, \ldots, x_n^*)$$

is a solution to the system where:

$$\begin{cases} f_1(x_1^*, x_2^*, \ldots, x_n^*) = 0 \\ f_2(x_1^*, x_2^*, \ldots, x_n^*) = 0 \\ \vdots \\ f_n(x_1^*, x_2^*, \ldots, x_n^*) = 0 \end{cases}$$

Equilibria Points

Equilibria points (or equilibrium points, critical points, stationary points) of a system of differential equations are points in the phase space where the system does not change over time.

Mathematically, for a system of differential equations:

$$\begin{cases} x_1' = f_1(x_1, x_2, \ldots, x_n) \\ x_2' = f_2(x_1, x_2, \ldots, x_n) \\ \vdots \\ x_n' = f_n(x_1, x_2, \ldots, x_n) \end{cases}$$

an equilibrium point

$$\mathbf{x}^* = (x_1^*, x_2^*, \ldots, x_n^*)$$

is a solution to the system where:

$$\begin{cases} f_1(x_1^*, x_2^*, \ldots, x_n^*) = 0 \\ f_2(x_1^*, x_2^*, \ldots, x_n^*) = 0 \\ \vdots \\ f_n(x_1^*, x_2^*, \ldots, x_n^*) = 0 \end{cases}$$

Equilibria Points

The intuition behind an equilibrium point in a system of differential equations can be understood as follows:

Equilibria points represent the states where the system is in balance, meaning that once the system reaches an equilibrium, it will stay there indefinitely if undisturbed.

Linear System:

$$\begin{cases} x_1' = x_2 \\ x_2' = -\frac{k}{m}x_1 - \frac{b}{m}x_2 \end{cases}$$

To find the equilibrium points, set $x_1' = 0$ and $x_2' = 0$:

$$x_2 = 0, \quad -\frac{k}{m}x_1 - \frac{b}{m}x_2 = 0.$$

This simplifies to $x_1 = 0$ and $x_2 = 0$.
Thus, the equilibrium point is $(0, 0)$.

Non-Linear System:

$$\begin{cases} x_1' = x_2 \\ x_2' = -\frac{g}{L}\sin(x_1) \end{cases}$$

Similarly, set $x_1' = 0$ and $x_2' = 0$:

$$x_2 = 0, \quad -\frac{g}{L}\sin(x_1) = 0.$$

This implies $\sin(x_1) = 0$, so $x_1 = \pi n$ where n is an integer.
Therefore, the equilibrium points are $(\pi n, 0)$.

Equilibria Points

The intuition behind an equilibrium point in a system of differential equations can be understood as follows:

Equilibria points represent the states where the system is in balance, meaning that once the system reaches an equilibrium, it will stay there indefinitely if undisturbed.

Linear System:

$$\begin{cases} x_1' = x_2 \\ x_2' = -\frac{k}{m}x_1 - \frac{b}{m}x_2 \end{cases}$$

To find the equilibrium points, set $x_1' = 0$ and $x_2' = 0$:

$$x_2 = 0, \quad -\frac{k}{m}x_1 - \frac{b}{m}x_2 = 0.$$

This simplifies to $x_1 = 0$ and $x_2 = 0$. Thus, the equilibrium point is $(0, 0)$.

Non-Linear System:

$$\begin{cases} x_1' = x_2 \\ x_2' = -\frac{g}{L}\sin(x_1) \end{cases}$$

Similarly, set $x_1' = 0$ and $x_2' = 0$:

$$x_2 = 0, \quad -\frac{g}{L}\sin(x_1) = 0.$$

This implies $\sin(x_1) = 0$, so $x_1 = \pi n$ where n is an integer.
Therefore, the equilibrium points are $(\pi n, 0)$.

Equilibria Points

The intuition behind an equilibrium point in a system of differential equations can be understood as follows:

Equilibria points represent the states where the system is in balance, meaning that once the system reaches an equilibrium, it will stay there indefinitely if undisturbed.

Linear System:

$$\begin{cases} x_1' = x_2 \\ x_2' = -\frac{k}{m}x_1 - \frac{b}{m}x_2 \end{cases}$$

To find the equilibrium points, set $x_1' = 0$ and $x_2' = 0$:

$$x_2 = 0, \quad -\frac{k}{m}x_1 - \frac{b}{m}x_2 = 0.$$

This simplifies to $x_1 = 0$ and $x_2 = 0$.
Thus, the equilibrium point is $(0, 0)$.

Non-Linear System:

$$\begin{cases} x_1' = x_2 \\ x_2' = -\frac{g}{L}\sin(x_1) \end{cases}$$

Similarly, set $x_1' = 0$ and $x_2' = 0$:

$$x_2 = 0, \quad -\frac{g}{L}\sin(x_1) = 0.$$

This implies $\sin(x_1) = 0$, so $x_1 = \pi n$ where n is an integer.
Therefore, the equilibrium points are $(\pi n, 0)$.

Equilibria Points

The intuition behind an equilibrium point in a system of differential equations can be understood as follows:

Equilibria points represent the states where the system is in balance, meaning that once the system reaches an equilibrium, it will stay there indefinitely if undisturbed.

Linear System:

$$\begin{cases} x_1' = x_2 \\ x_2' = -\frac{k}{m}x_1 - \frac{b}{m}x_2 \end{cases}$$

To find the equilibrium points, set $x_1' = 0$ and $x_2' = 0$:

$$x_2 = 0, \quad -\frac{k}{m}x_1 - \frac{b}{m}x_2 = 0.$$

This simplifies to $x_1 = 0$ and $x_2 = 0$. Thus, the equilibrium point is $(0, 0)$.

Non-Linear System:

$$\begin{cases} x_1' = x_2 \\ x_2' = -\frac{g}{L}\sin(x_1) \end{cases}$$

Similarly, set $x_1' = 0$ and $x_2' = 0$:

$$x_2 = 0, \quad -\frac{g}{L}\sin(x_1) = 0.$$

This implies $\sin(x_1) = 0$, so $x_1 = \pi n$ where n is an integer.
Therefore, the equilibrium points are $(\pi n, 0)$.

Equilibria Points

The intuition behind an equilibrium point in a system of differential equations can be understood as follows:

Equilibria points represent the states where the system is in balance, meaning that once the system reaches an equilibrium, it will stay there indefinitely if undisturbed.

Linear System:

$$\begin{cases} x_1' = x_2 \\ x_2' = -\frac{k}{m}x_1 - \frac{b}{m}x_2 \end{cases}$$

To find the equilibrium points, set $x_1' = 0$ and $x_2' = 0$:

$$x_2 = 0, \quad -\frac{k}{m}x_1 - \frac{b}{m}x_2 = 0.$$

This simplifies to $x_1 = 0$ and $x_2 = 0$.
Thus, the equilibrium point is $(0, 0)$.

Non-Linear System:

$$\begin{cases} x_1' = x_2 \\ x_2' = -\frac{g}{L}\sin(x_1) \end{cases}$$

Similarly, set $x_1' = 0$ and $x_2' = 0$:

$$x_2 = 0, \quad -\frac{g}{L}\sin(x_1) = 0.$$

This implies $\sin(x_1) = 0$, so $x_1 = \pi n$ where n is an integer.
Therefore, the equilibrium points are $(\pi n, 0)$.

Equilibria Points

The intuition behind an equilibrium point in a system of differential equations can be understood as follows:

Equilibria points represent the states where the system is in balance, meaning that once the system reaches an equilibrium, it will stay there indefinitely if undisturbed.

Linear System:

$$\begin{cases} x_1' = x_2 \\ x_2' = -\frac{k}{m}x_1 - \frac{b}{m}x_2 \end{cases}$$

To find the equilibrium points, set $x_1' = 0$ and $x_2' = 0$:

$$x_2 = 0, \quad -\frac{k}{m}x_1 - \frac{b}{m}x_2 = 0.$$

This simplifies to $x_1 = 0$ and $x_2 = 0$. Thus, the equilibrium point is $(0, 0)$.

Non-Linear System:

$$\begin{cases} x_1' = x_2 \\ x_2' = -\frac{g}{L}\sin(x_1) \end{cases}$$

Similarly, set $x_1' = 0$ and $x_2' = 0$:

$$x_2 = 0, \quad -\frac{g}{L}\sin(x_1) = 0.$$

This implies $\sin(x_1) = 0$, so $x_1 = \pi n$ where n is an integer.

Therefore, the equilibrium points are $(\pi n, 0)$.

Equilibria Points

The intuition behind an equilibrium point in a system of differential equations can be understood as follows:

Equilibria points represent the states where the system is in balance, meaning that once the system reaches an equilibrium, it will stay there indefinitely if undisturbed.

Linear System:

$$\begin{cases} x_1' = x_2 \\ x_2' = -\frac{k}{m}x_1 - \frac{b}{m}x_2 \end{cases}$$

To find the equilibrium points, set $x_1' = 0$ and $x_2' = 0$:

$$x_2 = 0, \quad -\frac{k}{m}x_1 - \frac{b}{m}x_2 = 0.$$

This simplifies to $x_1 = 0$ and $x_2 = 0$.
Thus, the equilibrium point is $(0, 0)$.

Non-Linear System:

$$\begin{cases} x_1' = x_2 \\ x_2' = -\frac{g}{L}\sin(x_1) \end{cases}$$

Similarly, set $x_1' = 0$ and $x_2' = 0$:

$$x_2 = 0, \quad -\frac{g}{L}\sin(x_1) = 0.$$

This implies $\sin(x_1) = 0$, so $x_1 = \pi n$ where n is an integer.
Therefore, the equilibrium points are $(\pi n, 0)$.

Equilibria Points

Damped harmonic oscillator

- The equilibrium point $(0, 0)$ represents the situation where the mass is at rest in its natural position (neither stretched nor compressed) and not moving;
- Physically, this is the point where all forces are balanced: the spring force (proportional to displacement) and the damping force (proportional to velocity) cancel each other out;
- If the system is at this equilibrium and not disturbed, it will stay at rest indefinitely because there are no net forces acting on it.

Equilibria Points

Damped harmonic oscillator

- The equilibrium point $(0, 0)$ represents the situation where the mass is at rest in its natural position (neither stretched nor compressed) and not moving;
- Physically, this is the point where all forces are balanced: the spring force (proportional to displacement) and the damping force (proportional to velocity) cancel each other out;
- If the system is at this equilibrium and not disturbed, it will stay at rest indefinitely because there are no net forces acting on it.

Equilibria Points

Damped harmonic oscillator

- The equilibrium point $(0, 0)$ represents the situation where the mass is at rest in its natural position (neither stretched nor compressed) and not moving;
- Physically, this is the point where all forces are balanced: the spring force (proportional to displacement) and the damping force (proportional to velocity) cancel each other out;
- If the system is at this equilibrium and not disturbed, it will stay at rest indefinitely because there are no net forces acting on it.

Equilibria Points

Simple pendulum:

- The equilibrium points correspond to specific angles $x_1 = \pi n$, where the pendulum is either hanging straight down (n even) or pointing straight up (n odd), and the pendulum is stationary ($x_2 = 0$);
  - For n even (e.g., $x_1 = 0$, $x_1 = 2\pi$), the pendulum is hanging straight down. This is a stable equilibrium because if you slightly push the pendulum, it will swing back to this downward position;
  - For n odd (e.g., $x_1 = \pi$), the pendulum is pointing straight up. This is an unstable equilibrium because a small disturbance will cause the pendulum to fall away from this position, eventually swinging downwards to the stable equilibrium.
- Thus, these equilibria represent the points where the pendulum has no kinetic energy (not moving) and potential energy is either minimized (stable, pendulum down) or maximized (unstable, pendulum up).

Equilibria Points

Simple pendulum:

- The equilibrium points correspond to specific angles $x_1 = \pi n$, where the pendulum is either hanging straight down (n even) or pointing straight up (n odd), and the pendulum is stationary ($x_2 = 0$);
  - For n even (e.g., $x_1 = 0$, $x_1 = 2\pi$), the pendulum is hanging straight down. This is a stable equilibrium because if you slightly push the pendulum, it will swing back to this downward position;
  - For n odd (e.g., $x_1 = \pi$), the pendulum is pointing straight up. This is an unstable equilibrium because a small disturbance will cause the pendulum to fall away from this position, eventually swinging downwards to the stable equilibrium.
- Thus, these equilibria represent the points where the pendulum has no kinetic energy (not moving) and potential energy is either minimized (stable, pendulum down) or maximized (unstable, pendulum up).

Equilibria Points

Simple pendulum:

- The equilibrium points correspond to specific angles $x_1 = \pi n$, where the pendulum is either hanging straight down (n even) or pointing straight up (n odd), and the pendulum is stationary ($x_2 = 0$);
  - For n even (e.g., $x_1 = 0$, $x_1 = 2\pi$), the pendulum is hanging straight down. This is a stable equilibrium because if you slightly push the pendulum, it will swing back to this downward position;
  - For n odd (e.g., $x_1 = \pi$), the pendulum is pointing straight up. This is an unstable equilibrium because a small disturbance will cause the pendulum to fall away from this position, eventually swinging downwards to the stable equilibrium.
- Thus, these equilibria represent the points where the pendulum has no kinetic energy (not moving) and potential energy is either minimized (stable, pendulum down) or maximized (unstable, pendulum up).

Equilibria Points

Simple pendulum:

- The equilibrium points correspond to specific angles $x_1 = \pi n$, where the pendulum is either hanging straight down (n even) or pointing straight up (n odd), and the pendulum is stationary ($x_2 = 0$);
  - For n even (e.g., $x_1 = 0$, $x_1 = 2\pi$), the pendulum is hanging straight down. This is a stable equilibrium because if you slightly push the pendulum, it will swing back to this downward position;
  - For n odd (e.g., $x_1 = \pi$), the pendulum is pointing straight up. This is an unstable equilibrium because a small disturbance will cause the pendulum to fall away from this position, eventually swinging downwards to the stable equilibrium.
- Thus, these equilibria represent the points where the pendulum has no kinetic energy (not moving) and potential energy is either minimized (stable, pendulum down) or maximized (unstable, pendulum up).

Stability Theory

Consider the system of DEs (in vector form):

$$\mathbf{X}' = \mathbf{f}(t, \mathbf{X})$$

with initial conditions:

$$\mathbf{X}(0) = \mathbf{X}_0. \tag{1}$$

Lyapunov Stability:

The solution $\varphi(t)$ of the initial value problem (IVP) (1) is said to be stable (in the sense of Lyapunov) if for any $\varepsilon > 0$, there exists $\delta = \delta(\varepsilon) > 0$ such that if

$$|\mathbf{X}(0) - \varphi(0)| < \delta,$$

then

$$|\mathbf{X}(t) - \varphi(t)| < \varepsilon$$

for all $t \geq 0$. Otherwise, the solution $\varphi(t)$ is said to be unstable.

Stability Theory

Consider the system of DEs (in vector form):

$$\mathbf{X}' = \mathbf{f}(t, \mathbf{X})$$

with initial conditions:

$$\mathbf{X}(0) = \mathbf{X}_0. \tag{1}$$

Lyapunov Stability:

The solution $\boldsymbol{\varphi}(t)$ of the initial value problem (IVP) (1) is said to be stable (in the sense of Lyapunov) if for any $\varepsilon > 0$, there exists $\delta = \delta(\varepsilon) > 0$ such that if

$$|\mathbf{X}(0) - \boldsymbol{\varphi}(0)| < \delta,$$

then

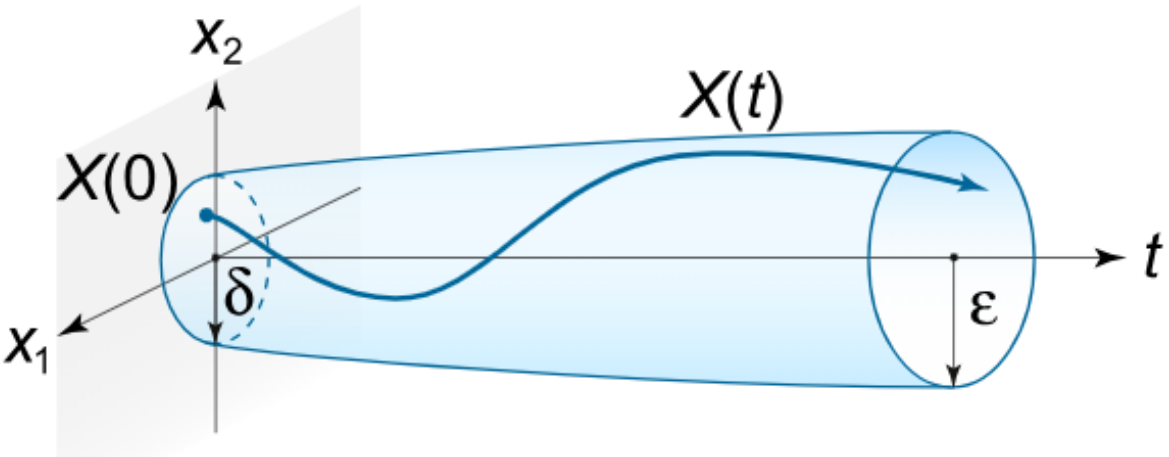
$$|\mathbf{X}(t) - \boldsymbol{\varphi}(t)| < \varepsilon$$

for all $t \geq 0$. Otherwise, the solution $\boldsymbol{\varphi}(t)$ is said to be unstable.

Stability in the sense of Lyapunov

In the case $n = 2$

Lyapunov stability means that any trajectory $\mathbf{X}(t)$, which starts at $\delta(\varepsilon)$-neighborhood of the point $\boldsymbol{\varphi}(0)$, remains inside the tube with a maximum radius ε for all $t \geq 0$.



Stability Theory

Asymptotic Stability:

If the solution $\boldsymbol{\varphi}(t)$ of the IVP (1) is not only stable in the sense of Lyapunov, but also satisfies the relationship

$$\lim_{t\to\infty} |\mathbf{X}(t) - \boldsymbol{\varphi}(t)| = 0,$$

provided that

$$|\mathbf{X}(0) - \boldsymbol{\varphi}(0)| < \delta,$$

then we say that the solution $\boldsymbol{\varphi}(t)$ is asymptotically stable.

Asymptotic Stability

In the case $n = 2$

All solutions that are sufficiently close to $\varphi(0)$ at the initial time, gradually converge to $\varphi(t)$ with increasing t.

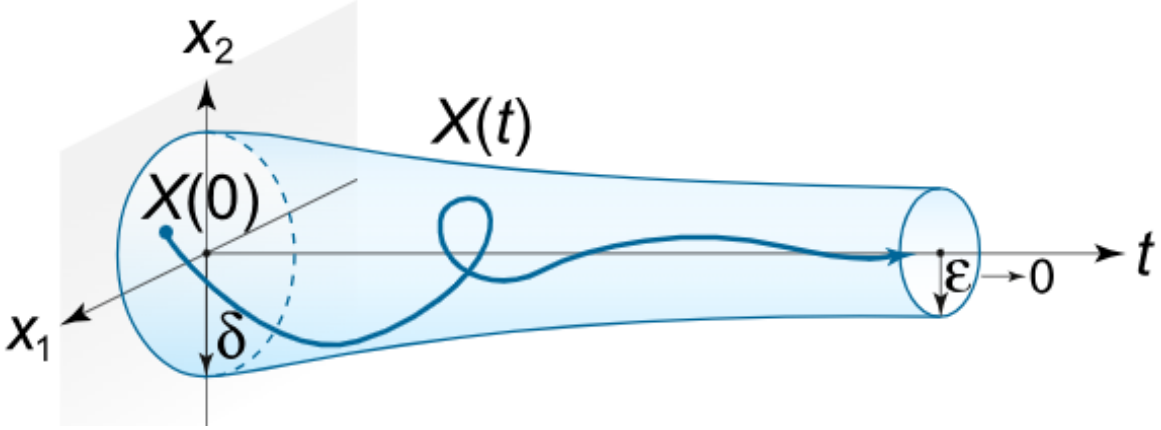


Table of Contents

1. 4.1. Basic theory of systems of ODEs
2. 4.2. Equilibria Points and Basic Concepts of Stability Theory
3. 4.3. Equilibrium Points of Linear Autonomous Systems and its Stability

Matrix form of Linear Systems

When the equations in the system are linear, matrix algebra provides a compact notation for expressing such systems. Moreover, the notation itself suggests the new ways of characterizing the solution properties.

Introduce the following column vectors and a matrix with entries which are the elements of the linear system:

$$\mathbf{x} = \begin{bmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{bmatrix}, \quad \mathbf{x}' = \begin{bmatrix} x_1'(t) \\ \vdots \\ x_n'(t) \end{bmatrix}, \quad \mathbf{f} = \begin{bmatrix} f_1(t) \\ \vdots \\ f_n(t) \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} a_{11}(t) & \cdots & a_{1n}(t) \\ \vdots & \ddots & \vdots \\ a_{n1}(t) & \cdots & a_{nn}(t) \end{bmatrix}.$$

Then the linear system can be written in the compact matrix form:

$$\mathbf{x}' = \mathbf{A}\mathbf{x} + \mathbf{f} \tag{2}$$

Matrix form of Linear Systems

When the equations in the system are linear, matrix algebra provides a compact notation for expressing such systems. Moreover, the notation itself suggests the new ways of characterizing the solution properties.

Introduce the following column vectors and a matrix with entries which are the elements of the linear system:

$$\mathbf{x} = \begin{bmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{bmatrix}, \quad \mathbf{x}' = \begin{bmatrix} x_1'(t) \\ \vdots \\ x_n'(t) \end{bmatrix}, \quad \mathbf{f} = \begin{bmatrix} f_1(t) \\ \vdots \\ f_n(t) \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} a_{11}(t) & \cdots & a_{1n}(t) \\ \vdots & \ddots & \vdots \\ a_{n1}(t) & \cdots & a_{nn}(t) \end{bmatrix}.$$

Then the linear system can be written in the compact matrix form:

$$\mathbf{x}' = \mathbf{A}\mathbf{x} + \mathbf{f} \tag{2}$$

Matrix form of Linear Systems

We say that a system of n linear differential equations is in normal form if it is expressed as

$$\mathbf{x}' = \mathbf{A}\mathbf{x} + \mathbf{f},$$

where $\mathbf{x} = \mathrm{col}(x_1(t), \ldots, x_n(t))$, $\mathbf{f} = \mathrm{col}(f_1(t), \ldots, f_n(t))$, and $\mathbf{A} = [a_{ij}(t)]$ is an $n \times n$ matrix.

As with a scalar linear differential equation, a system is called homogeneous when $\mathbf{f} = 0$; otherwise, it is called nonhomogeneous.

When the elements of $\mathbf{A}$ are all constants, the system is said to have constant coefficients.

Matrix form of Linear Systems

We say that a system of n linear differential equations is in normal form if it is expressed as

$$\mathbf{x}' = \mathbf{A}\mathbf{x} + \mathbf{f},$$

where $\mathbf{x} = \mathrm{col}(x_1(t), \ldots, x_n(t))$, $\mathbf{f} = \mathrm{col}(f_1(t), \ldots, f_n(t))$, and $\mathbf{A} = [a_{ij}(t)]$ is an $n \times n$ matrix.

As with a scalar linear differential equation, a system is called homogeneous when $\mathbf{f} = 0$; otherwise, it is called nonhomogeneous.

When the elements of $\mathbf{A}$ are all constants, the system is said to have constant coefficients.

Matrix form of Linear Systems

We say that a system of n linear differential equations is in normal form if it is expressed as

$$\mathbf{x}' = \mathbf{A}\mathbf{x} + \mathbf{f},$$

where $\mathbf{x} = \mathrm{col}(x_1(t), \ldots, x_n(t))$, $\mathbf{f} = \mathrm{col}(f_1(t), \ldots, f_n(t))$, and $\mathbf{A} = [a_{ij}(t)]$ is an $n \times n$ matrix.

As with a scalar linear differential equation, a system is called homogeneous when $\mathbf{f} = 0$; otherwise, it is called nonhomogeneous.

When the elements of $\mathbf{A}$ are all constants, the system is said to have constant coefficients.

Homogeneous Linear Systems with Constant Coefficients

Let a first order linear homogeneous system with constant coefficients be given:

$$\begin{cases} \frac{dx}{dt} = a_{11}x + a_{12}y \\ \frac{dy}{dt} = a_{21}x + a_{22}y \end{cases}.$$

This system of equations is autonomous since the right hand sides of the equations do not explicitly contain the independent variable t.

The equilibrium positions can be found by solving the equation:

$$A\mathbf{X} = \mathbf{0}.$$

This equation has

- the unique solution $\mathbf{X} = \mathbf{0}$ if the matrix A is nonsingular, i.e. provided that $\det A \neq 0$;
- an infinite number of equilibrium points if the matrix A is a singular matrix;

After we found the equilibrium, we need to classify it. Classification of equilibrium points is determined by the eigenvalues λ_1, λ_2 of the matrix A. And, thus, the problem turns into eigenvalue problem.

Homogeneous Linear Systems with Constant Coefficients

Let a first order linear homogeneous system with constant coefficients be given:

$$\begin{cases} \frac{dx}{dt} = a_{11}x + a_{12}y \\ \frac{dy}{dt} = a_{21}x + a_{22}y \end{cases}.$$

This system of equations is autonomous since the right hand sides of the equations do not explicitly contain the independent variable t.

The equilibrium positions can be found by solving the equation:

$$A\mathbf{X} = \mathbf{0}.$$

This equation has

- the unique solution $\mathbf{X} = \mathbf{0}$ if the matrix A is nonsingular, i.e. provided that $\det A \neq 0$;
- an infinite number of equilibrium points if the matrix A is a singular matrix;

After we found the equilibrium, we need to classify it. Classification of equilibrium points is determined by the eigenvalues λ_1, λ_2 of the matrix A. And, thus, the problem turns into eigenvalue problem.

Homogeneous Linear Systems with Constant Coefficients

Let a first order linear homogeneous system with constant coefficients be given:

$$\begin{cases} \frac{dx}{dt} = a_{11}x + a_{12}y \\ \frac{dy}{dt} = a_{21}x + a_{22}y \end{cases}.$$

This system of equations is autonomous since the right hand sides of the equations do not explicitly contain the independent variable t.

The equilibrium positions can be found by solving the equation:

$$A\mathbf{X} = \mathbf{0}.$$

This equation has

- the unique solution $\mathbf{X} = \mathbf{0}$ if the matrix A is nonsingular, i.e. provided that $\det A \neq 0$;
- an infinite number of equilibrium points if the matrix A is a singular matrix;

After we found the equilibrium, we need to classify it. Classification of equilibrium points is determined by the eigenvalues λ_1, λ_2 of the matrix A. And, thus, the problem turns into eigenvalue problem.

Homogeneous Linear Systems with Constant Coefficients

Let a first order linear homogeneous system with constant coefficients be given:

$$\begin{cases} \frac{dx}{dt} = a_{11}x + a_{12}y \\ \frac{dy}{dt} = a_{21}x + a_{22}y \end{cases}.$$

This system of equations is autonomous since the right hand sides of the equations do not explicitly contain the independent variable t.

The equilibrium positions can be found by solving the equation:

$$A\mathbf{X} = \mathbf{0}.$$

This equation has

- the unique solution $\mathbf{X} = \mathbf{0}$ if the matrix A is nonsingular, i.e. provided that $\det A \neq 0$;
- an infinite number of equilibrium points if the matrix A is a singular matrix;

After we found the equilibrium, we need to classify it. Classification of equilibrium points is determined by the eigenvalues λ_1, λ_2 of the matrix A. And, thus, the problem turns into eigenvalue problem.

Homogeneous Linear Systems with Constant Coefficients

Let a first order linear homogeneous system with constant coefficients be given:

$$\begin{cases} \frac{dx}{dt} = a_{11}x + a_{12}y \\ \frac{dy}{dt} = a_{21}x + a_{22}y \end{cases}.$$

This system of equations is autonomous since the right hand sides of the equations do not explicitly contain the independent variable t.

The equilibrium positions can be found by solving the equation:

$$A\mathbf{X} = \mathbf{0}.$$

This equation has

- the unique solution $\mathbf{X} = \mathbf{0}$ if the matrix A is nonsingular, i.e. provided that $\det A \neq 0$;
- an infinite number of equilibrium points if the matrix A is a singular matrix;

After we found the equilibrium, we need to classify it. Classification of equilibrium points is determined by the eigenvalues λ_1, λ_2 of the matrix A. And, thus, the problem turns into eigenvalue problem.

Homogeneous Linear Systems with Constant Coefficients

Let a first order linear homogeneous system with constant coefficients be given:

$$\begin{cases} \frac{dx}{dt} = a_{11}x + a_{12}y \\ \frac{dy}{dt} = a_{21}x + a_{22}y \end{cases}.$$

This system of equations is autonomous since the right hand sides of the equations do not explicitly contain the independent variable t.

The equilibrium positions can be found by solving the equation:

$$A\mathbf{X} = \mathbf{0}.$$

This equation has

- the unique solution $\mathbf{X} = \mathbf{0}$ if the matrix A is nonsingular, i.e. provided that $\det A \neq 0$;
- an infinite number of equilibrium points if the matrix A is a singular matrix;

After we found the equilibrium, we need to classify it. Classification of equilibrium points is determined by the eigenvalues λ_1, λ_2 of the matrix A. And, thus, the problem turns into eigenvalue problem.

Types of Equilibrium Points

In general, when the matrix A is nonsingular, there are 4 different types of equilibrium points:

	Equilibrium point	Eigenvalues λ_1, λ_2
1	Node	λ_1, λ_2 are real numbers of the same sign
2	Saddle	λ_1, λ_2 are real numbers and non-zero of the opposite sign
3	Focus	λ_1, λ_2 are complex numbers; the real parts are equal and non-zero
4	Center	λ_1, λ_2 are purely imaginary numbers

The stability of equilibrium points is determined by the following theorems on stability.

Types of Equilibrium Points

In general, when the matrix A is nonsingular, there are 4 different types of equilibrium points:

	Equilibrium point	Eigenvalues λ_1, λ_2
1	Node	λ_1, λ_2 are real numbers of the same sign
2	Saddle	λ_1, λ_2 are real numbers and non-zero of the opposite sign
3	Focus	λ_1, λ_2 are complex numbers; the real parts are equal and non-zero
4	Center	λ_1, λ_2 are purely imaginary numbers

The stability of equilibrium points is determined by the following theorems on stability.

Stability of Linear Systems

The linear system

$$\mathbf{X}' = A(t)\,\mathbf{X} + \mathbf{f}(t)$$

is said to be stable if all its solutions are stable in the sense of Lyapunov.

The non-homogeneous linear system is stable with any free term $\mathbf{f}(t)$ if the zero solution of the associated homogeneous system

$$\mathbf{X}' = A(t)\,\mathbf{X}$$

is stable.

Therefore, when investigating stability in the class of linear systems, it is sufficient to analyze the homogeneous differential systems.

Stability of Linear Systems

The linear system

$$\mathbf{X}' = A(t)\,\mathbf{X} + \mathbf{f}(t)$$

is said to be stable if all its solutions are stable in the sense of Lyapunov.

The non-homogeneous linear system is stable with any free term $\mathbf{f}(t)$ if the zero solution of the associated homogeneous system

$$\mathbf{X}' = A(t)\,\mathbf{X}$$

is stable.

Therefore, when investigating stability in the class of linear systems, it is sufficient to analyze the homogeneous differential systems.

Stability of Linear Systems

The linear system

$$\mathbf{X}' = A(t)\,\mathbf{X} + \mathbf{f}(t)$$

is said to be stable if all its solutions are stable in the sense of Lyapunov.

The non-homogeneous linear system is stable with any free term $\mathbf{f}(t)$ if the zero solution of the associated homogeneous system

$$\mathbf{X}' = A(t)\,\mathbf{X}$$

is stable.

Therefore, when investigating stability in the class of linear systems, it is sufficient to analyze the homogeneous differential systems.

Stability of Linear Systems

Consider the homogeneous linear system

$$\mathbf{X}' = A\mathbf{X},$$

where A is a constant matrix of size $n \times n$.
Such a system (which is also autonomous) has the zero solution

$$\mathbf{X}(t) = \mathbf{0}.$$

The stability of this solution is determined by the following theorems.

Theorem 1 (Lyapunov Stability):

A linear homogeneous system with constant coefficients is stable in the sense of Lyapunov if and only if all eigenvalues λ_i of A satisfy the condition

$$\operatorname{Re}[\lambda_i] \leq 0 \quad (i = 1, 2, \ldots, n),$$

If the real part of an eigenvalue is equal to zero, the algebraic and geometric multiplicity of the eigenvalue must be the same (i.e. the corresponding Jordan block must be of size 1×1).

Stability of Linear Systems

Consider the homogeneous linear system

$$\mathbf{X}' = A\mathbf{X},$$

where A is a constant matrix of size $n \times n$.
Such a system (which is also autonomous) has the zero solution

$$\mathbf{X}(t) = \mathbf{0}.$$

The stability of this solution is determined by the following theorems.

Theorem 1 (Lyapunov Stability):

A linear homogeneous system with constant coefficients is stable in the sense of Lyapunov if and only if all eigenvalues λ_i of A satisfy the condition

$$\operatorname{Re}[\lambda_i] \leq 0 \quad (i = 1, 2, \ldots, n),$$

If the real part of an eigenvalue is equal to zero, the algebraic and geometric multiplicity of the eigenvalue must be the same (i.e. the corresponding Jordan block must be of size 1×1).

Stability of Linear Systems

Consider the homogeneous linear system

$$\mathbf{X}' = A\mathbf{X},$$

where A is a constant matrix of size $n \times n$.
Such a system (which is also autonomous) has the zero solution

$$\mathbf{X}(t) = \mathbf{0}.$$

The stability of this solution is determined by the following theorems.

Theorem 1 (Lyapunov Stability):

A linear homogeneous system with constant coefficients is stable in the sense of Lyapunov if and only if all eigenvalues λ_i of A satisfy the condition

$$\operatorname{Re}[\lambda_i] \leq 0 \quad (i = 1, 2, \ldots, n),$$

If the real part of an eigenvalue is equal to zero, the algebraic and geometric multiplicity of the eigenvalue must be the same (i.e. the corresponding Jordan block must be of size 1×1).

Stability of Linear Systems

Consider the homogeneous linear system

$$\mathbf{X}' = A\mathbf{X},$$

where A is a constant matrix of size $n \times n$.
Such a system (which is also autonomous) has the zero solution

$$\mathbf{X}(t) = \mathbf{0}.$$

The stability of this solution is determined by the following theorems.

Theorem 1 (Lyapunov Stability):

A linear homogeneous system with constant coefficients is stable in the sense of Lyapunov if and only if all eigenvalues λ_i of A satisfy the condition

$$\operatorname{Re}[\lambda_i] \leq 0 \quad (i = 1, 2, \ldots, n),$$

If the real part of an eigenvalue is equal to zero, the algebraic and geometric multiplicity of the eigenvalue must be the same (i.e. the corresponding Jordan block must be of size 1×1).

Stability of Linear Systems

Theorem 2 (Asymptotic Stability):

A linear homogeneous system with constant coefficients is asymptotically stable if and only if all eigenvalues λ_i have negative real parts:

$$\mathrm{Re}\,[\lambda_i] < 0 \quad (i = 1, 2, \ldots, n)$$

Theorem 3 (Instability):

A linear homogeneous system with constant coefficients is unstable if at least one of the conditions is satisfied:

- The matrix A has an eigenvalue λ_i with a positive real part;
- The matrix A has an eigenvalue λ_i with zero real part, and the geometric multiplicity of the eigenvalue λ_i is less than its algebraic multiplicity;

Stability of Linear Systems

Theorem 2 (Asymptotic Stability):

A linear homogeneous system with constant coefficients is asymptotically stable if and only if all eigenvalues λ_i have negative real parts:

$$\mathrm{Re}\,[\lambda_i] < 0 \quad (i = 1, 2, \ldots, n)$$

Theorem 3 (Instability):

A linear homogeneous system with constant coefficients is unstable if at least one of the conditions is satisfied:

- The matrix A has an eigenvalue λ_i with a positive real part;
- The matrix A has an eigenvalue λ_i with zero real part, and the geometric multiplicity of the eigenvalue λ_i is less than its algebraic multiplicity;

Phase portraits

How to Sketch a Phase Portrait

1. Find the eigenvalues of the matrix;
2. Determine the type of the equilibrium point and the character of stability;
3. Find the equations of the isoclines:

$$\frac{dx}{dt} = a_{11}x + a_{12}y \quad \text{(vertical isocline)}\,,$$

$$\frac{dy}{dt} = a_{21}x + a_{22}y \quad \text{(horizontal isocline)}\,.$$

4. If the equilibrium position is a node or a saddle, it is necessary to compute the eigenvectors and draw the asymptotes parallel to the eigenvectors and passing through the origin;
5. Schematically draw the phase portrait;
6. Show the direction of motion along the phase trajectories. In the case of a focus or center, one should determine the direction of trajectories twisting.

Example 1

Investigate the equilibrium positions of the linear autonomous system and draw its phase portrait:

$$\begin{cases} \frac{dx_1}{dt} = -x_1 \\ \frac{dx_2}{dt} = 2x_1 - 2x_2 \end{cases}$$

Step 1

We write the matrix of the system and compute its determinant:

$$A = \begin{bmatrix} -1 & 0 \\ 2 & -2 \end{bmatrix}, \quad \det A = \begin{vmatrix} -1 & 0 \\ 2 & -2 \end{vmatrix} = 2 \neq 0.$$

As $\det A \neq 0$, the system has the unique equilibrium point

$$\mathbf{X} = \mathbf{0}.$$

We find the eigenvalues of the matrix A: $\det(A - \lambda I) = 0, \Rightarrow$

$$\begin{vmatrix} -1-\lambda & 0 \\ 2 & -2-\lambda \end{vmatrix} = 0, \quad \Rightarrow (\lambda+1)(\lambda+2) = 0, \quad \Rightarrow \lambda_1 = -1,\ \lambda_2 = -2.$$

Example 1

Investigate the equilibrium positions of the linear autonomous system and draw its phase portrait:

$$\begin{cases} \frac{dx_1}{dt} = -x_1 \\ \frac{dx_2}{dt} = 2x_1 - 2x_2 \end{cases}$$

Step 1

We write the matrix of the system and compute its determinant:

$$A = \begin{bmatrix} -1 & 0 \\ 2 & -2 \end{bmatrix}, \quad \det A = \begin{vmatrix} -1 & 0 \\ 2 & -2 \end{vmatrix} = 2 \neq 0.$$

As $\det A \neq 0$, the system has the unique equilibrium point

$$\mathbf{X} = \mathbf{0}.$$

We find the eigenvalues of the matrix A: $\det(A - \lambda I) = 0, \Rightarrow$

$$\begin{vmatrix} -1-\lambda & 0 \\ 2 & -2-\lambda \end{vmatrix} = 0, \quad \Rightarrow (\lambda+1)(\lambda+2) = 0, \quad \Rightarrow \lambda_1 = -1,\ \lambda_2 = -2.$$

Example 1

Investigate the equilibrium positions of the linear autonomous system and draw its phase portrait:

$$\begin{cases} \frac{dx_1}{dt} = -x_1 \\ \frac{dx_2}{dt} = 2x_1 - 2x_2 \end{cases}$$

Step 1

We write the matrix of the system and compute its determinant:

$$A = \begin{bmatrix} -1 & 0 \\ 2 & -2 \end{bmatrix}, \quad \det A = \begin{vmatrix} -1 & 0 \\ 2 & -2 \end{vmatrix} = 2 \neq 0.$$

As $\det A \neq 0$, the system has the unique equilibrium point

$$\mathbf{X} = \mathbf{0}.$$

We find the eigenvalues of the matrix A: $\det(A - \lambda I) = 0, \Rightarrow$

$$\begin{vmatrix} -1-\lambda & 0 \\ 2 & -2-\lambda \end{vmatrix} = 0, \quad \Rightarrow (\lambda+1)(\lambda+2) = 0, \quad \Rightarrow \lambda_1 = -1,\ \lambda_2 = -2.$$

Example 1

Investigate the equilibrium positions of the linear autonomous system and draw its phase portrait:

$$\begin{cases} \frac{dx_1}{dt} = -x_1 \\ \frac{dx_2}{dt} = 2x_1 - 2x_2 \end{cases}$$

Step 1

We write the matrix of the system and compute its determinant:

$$A = \begin{bmatrix} -1 & 0 \\ 2 & -2 \end{bmatrix}, \quad \det A = \begin{vmatrix} -1 & 0 \\ 2 & -2 \end{vmatrix} = 2 \neq 0.$$

As $\det A \neq 0$, the system has the unique equilibrium point

$$\mathbf{X} = \mathbf{0}.$$

We find the eigenvalues of the matrix A: $\det(A - \lambda I) = 0, \Rightarrow$

$$\begin{vmatrix} -1-\lambda & 0 \\ 2 & -2-\lambda \end{vmatrix} = 0, \quad \Rightarrow (\lambda+1)(\lambda+2) = 0, \quad \Rightarrow \lambda_1 = -1,\ \lambda_2 = -2.$$

Example 1

Investigate the equilibrium positions of the linear autonomous system and draw its phase portrait:

$$\begin{cases} \frac{dx_1}{dt} = -x_1 \\ \frac{dx_2}{dt} = 2x_1 - 2x_2 \end{cases}$$

Step 2

Both eigenvalues are real and negative, so the equilibrium point $\mathbf{X} = \mathbf{0}$ is a stable node.

Example 1

Investigate the equilibrium positions of the linear autonomous system and draw its phase portrait:

$$\begin{cases} \frac{dx_1}{dt} = -x_1 \\ \frac{dx_2}{dt} = 2x_1 - 2x_2 \end{cases}$$

Step 3

We derive the isocline equations, i.e. the lines which are tangent to the phase trajectories. The vertical isocline is given by

$$\frac{dx}{dt} = -x = 0 \text{ or } x = 0.$$

The horizontal isocline is written as

$$\frac{dy}{dt} = 2x - 2y = 0 \text{ or } y = x.$$

Example 1

Investigate the equilibrium positions of the linear autonomous system and draw its phase portrait:

$$\begin{cases} \frac{dx_1}{dt} = -x_1 \\ \frac{dx_2}{dt} = 2x_1 - 2x_2 \end{cases}$$

Step 4

Find the equations of the asymptotes. This can be done by calculating the eigenvectors $\mathbf{V}_1, \mathbf{V}_2$ of the matrix A:

$$(A - \lambda_1 I)\,\mathbf{V}_1 = \mathbf{0}, \;\Rightarrow \begin{bmatrix} -1+1 & 0 \\ 2 & -2+1 \end{bmatrix} \begin{bmatrix} V_{11} \\ V_{21} \end{bmatrix} = \mathbf{0}, \;\Rightarrow \begin{bmatrix} 0 & 0 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} V_{11} \\ V_{21} \end{bmatrix} = \mathbf{0}$$

$$\Rightarrow 2V_{11} - V_{21} = 0, \;\Rightarrow V_{11} = 1,\; V_{21} = 2, \;\Rightarrow \mathbf{V}_1 = \begin{bmatrix} V_{11} \\ V_{21} \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix};$$

Example 1

Investigate the equilibrium positions of the linear autonomous system and draw its phase portrait:

$$\begin{cases} \frac{dx_1}{dt} = -x_1 \\ \frac{dx_2}{dt} = 2x_1 - 2x_2 \end{cases}$$

Step 4

Find the equations of the asymptotes. This can be done by calculating the eigenvectors $\mathbf{V}_1, \mathbf{V}_2$ of the matrix A:

$$(A - \lambda_2 I)\,\mathbf{V}_2 = \mathbf{0}, \;\Rightarrow \begin{bmatrix} -1+2 & 0 \\ 2 & -2+2 \end{bmatrix} \begin{bmatrix} V_{12} \\ V_{22} \end{bmatrix} = \mathbf{0}, \Rightarrow \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} V_{12} \\ V_{22} \end{bmatrix} = \mathbf{0}$$

$$\Rightarrow \begin{cases} 1 \cdot V_{12} + 0 \cdot V_{22} = 0 \\ 2 \cdot V_{12} + 0 \cdot V_{22} = 0 \end{cases}, \;\Rightarrow V_{12} = 0,\; V_{22} = 1, \;\Rightarrow \mathbf{V}_2 = \begin{bmatrix} V_{12} \\ V_{22} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Example 1

Investigate the equilibrium positions of the linear autonomous system and draw its phase portrait:

$$\begin{cases} \frac{dx_1}{dt} = -x_1 \\ \frac{dx_2}{dt} = 2x_1 - 2x_2 \end{cases}$$

Step 5

- Draw an x_1x_2-plane;
- Show the eigenvectors $\mathbf{V}_1, \mathbf{V}_2$;
- Show the horizontal isocline $y = x$;
- Sketch the phase portrait of the system

Example 1

Investigate the equilibrium positions of the linear autonomous system and draw its phase portrait:

$$\begin{cases} \frac{dx_1}{dt} = -x_1 \\ \frac{dx_2}{dt} = 2x_1 - 2x_2 \end{cases}$$

Step 5

- Draw an x_1x_2-plane;
- Show the eigenvectors $\mathbf{V}_1, \mathbf{V}_2$;
- Show the horizontal isocline $y = x$;
- Sketch the phase portrait of the system

Example 1

Investigate the equilibrium positions of the linear autonomous system and draw its phase portrait:

$$\begin{cases} \frac{dx_1}{dt} = -x_1 \\ \frac{dx_2}{dt} = 2x_1 - 2x_2 \end{cases}$$

Step 5

- Draw an x_1x_2-plane;
- Show the eigenvectors $\mathbf{V}_1, \mathbf{V}_2$;
- Show the horizontal isocline $y = x$;
- Sketch the phase portrait of the system

Example 1

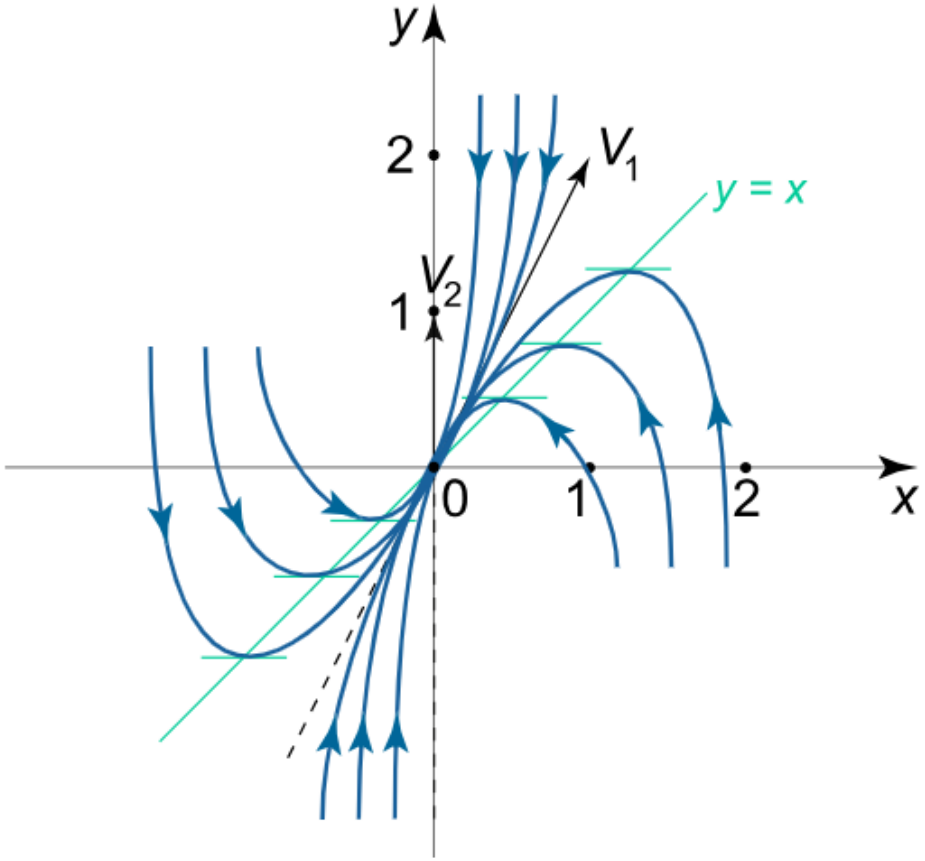
Investigate the equilibrium positions of the linear autonomous system and draw its phase portrait:

$$\begin{cases} \frac{dx_1}{dt} = -x_1 \\ \frac{dx_2}{dt} = 2x_1 - 2x_2 \end{cases}$$

Step 5

- Draw an x_1x_2-plane;
- Show the eigenvectors $\mathbf{V}_1, \mathbf{V}_2$;
- Show the horizontal isocline $y = x$;
- Sketch the phase portrait of the system

Example 1



The phase trajectories approach zero touching the line directed along the vector $\mathbf{V}_1$, as this eigenvector corresponds to the smallest (in absolute value) eigenvalue: $|\lambda_1| = 1$.

Example 2

Investigate the equilibrium positions of the dynamic system and sketch its phase portrait:

$$\begin{cases} \frac{dx_1}{dt} = x_1 + 3x_2 \\ \frac{dx_2}{dt} = 2x_1 \end{cases}$$

Step 1

We write the matrix of the system and compute its determinant:

$$A = \begin{pmatrix} 1 & 3 \\ 2 & 0 \end{pmatrix}, \quad \det A = \begin{vmatrix} 1 & 3 \\ 2 & 0 \end{vmatrix} = -6 \neq 0.$$

As $\det A \neq 0$, the system has the unique equilibrium point

$$\mathbf{X} = \mathbf{0}.$$

We find the eigenvalues of the matrix A: $\det(A - \lambda I) = 0, \Rightarrow$

$$\begin{vmatrix} 1-\lambda & 3 \\ 2 & -\lambda \end{vmatrix} = 0, \quad \Rightarrow \lambda^2 - \lambda + 6 = 0, \quad \Rightarrow \lambda_1 = 3,\ \lambda_2 = -2.$$

Example 2

Investigate the equilibrium positions of the dynamic system and sketch its phase portrait:

$$\begin{cases} \frac{dx_1}{dt} = x_1 + 3x_2 \\ \frac{dx_2}{dt} = 2x_1 \end{cases}$$

Step 1

We write the matrix of the system and compute its determinant:

$$A = \begin{pmatrix} 1 & 3 \\ 2 & 0 \end{pmatrix}, \quad \det A = \begin{vmatrix} 1 & 3 \\ 2 & 0 \end{vmatrix} = -6 \neq 0.$$

As $\det A \neq 0$, the system has the unique equilibrium point

$$\mathbf{X} = \mathbf{0}.$$

We find the eigenvalues of the matrix A: $\det(A - \lambda I) = 0, \Rightarrow$

$$\begin{vmatrix} 1-\lambda & 3 \\ 2 & -\lambda \end{vmatrix} = 0, \quad \Rightarrow \lambda^2 - \lambda + 6 = 0, \quad \Rightarrow \lambda_1 = 3,\ \lambda_2 = -2.$$

Example 2

Investigate the equilibrium positions of the dynamic system and sketch its phase portrait:

$$\begin{cases} \frac{dx_1}{dt} = x_1 + 3x_2 \\ \frac{dx_2}{dt} = 2x_1 \end{cases}$$

Step 1

We write the matrix of the system and compute its determinant:

$$A = \begin{pmatrix} 1 & 3 \\ 2 & 0 \end{pmatrix}, \quad \det A = \begin{vmatrix} 1 & 3 \\ 2 & 0 \end{vmatrix} = -6 \neq 0.$$

As $\det A \neq 0$, the system has the unique equilibrium point

$$\mathbf{X} = \mathbf{0}.$$

We find the eigenvalues of the matrix A: $\det(A - \lambda I) = 0, \Rightarrow$

$$\begin{vmatrix} 1-\lambda & 3 \\ 2 & -\lambda \end{vmatrix} = 0, \quad \Rightarrow \lambda^2 - \lambda + 6 = 0, \quad \Rightarrow \lambda_1 = 3,\ \lambda_2 = -2.$$

Example 2

Investigate the equilibrium positions of the dynamic system and sketch its phase portrait:

$$\begin{cases} \frac{dx_1}{dt} = x_1 + 3x_2 \\ \frac{dx_2}{dt} = 2x_1 \end{cases}$$

Step 1

We write the matrix of the system and compute its determinant:

$$A = \begin{pmatrix} 1 & 3 \\ 2 & 0 \end{pmatrix}, \quad \det A = \begin{vmatrix} 1 & 3 \\ 2 & 0 \end{vmatrix} = -6 \neq 0.$$

As $\det A \neq 0$, the system has the unique equilibrium point

$$\mathbf{X} = \mathbf{0}.$$

We find the eigenvalues of the matrix A: $\det(A - \lambda I) = 0, \Rightarrow$

$$\begin{vmatrix} 1-\lambda & 3 \\ 2 & -\lambda \end{vmatrix} = 0, \quad \Rightarrow \lambda^2 - \lambda + 6 = 0, \quad \Rightarrow \lambda_1 = 3,\ \lambda_2 = -2.$$

Example 2

Investigate the equilibrium positions of the linear autonomous system and draw its phase portrait:

$$\begin{cases} \frac{dx_1}{dt} = x_1 + 3x_2 \\ \frac{dx_2}{dt} = 2x_1 \end{cases}$$

Step 2

Both eigenvalues are real and opposite sign, the zero equilibrium point is a saddle.

Example 2

Investigate the equilibrium positions of the linear autonomous system and draw its phase portrait:

$$\begin{cases} \frac{dx_1}{dt} = x_1 + 3x_2 \\ \frac{dx_2}{dt} = 2x_1 \end{cases}$$

Step 3

Define the equations of isoclines. The vertical isocline is described by the linear function:

$$\frac{dx}{dt} = x + 3y = 0, \ \Rightarrow y = -\frac{x}{3}.$$

The equation of the horizontal isocline is

$$\frac{dy}{dt} = 2x = 0, \ \Rightarrow x = 0 \ (y - \text{axis}).$$

Example 2

Investigate the equilibrium positions of the linear autonomous system and draw its phase portrait:

$$\begin{cases} \frac{dx_1}{dt} = x_1 + 3x_2 \\ \frac{dx_2}{dt} = 2x_1 \end{cases}$$

Step 4

Find the equations of the separatrices which have the form $y = kx$. Substituting this into the original system, we obtain a quadratic equation for the coefficient k:

$$\begin{cases} \frac{dx}{dt} = x + 3y \\ \frac{dy}{dt} = 2x \end{cases}, \quad \Rightarrow \begin{cases} \frac{dx}{dt} = x + 3kx \\ \frac{kdx}{dt} = 2x \end{cases},$$

$$\Rightarrow 2x = k\left(x + 3kx\right), \ \Rightarrow 3k^2x + kx - 2x = 0, \ \Rightarrow 3k^2 + k - 2 = 0,$$

$$\Rightarrow D = 24, \ k_{1,2} = \frac{-1 \pm 5}{6} = -1, \ \frac{2}{3}.$$

Example 2

Investigate the equilibrium positions of the linear autonomous system and draw its phase portrait:

$$\begin{cases} \frac{dx_1}{dt} = x_1 + 3x_2 \\ \frac{dx_2}{dt} = 2x_1 \end{cases}$$

Step 5

Draw the isoclines and separatrices on the phase plane and sketch the phase trajectories

Determine the direction of motion along the phase trajectories.

Take, for example, the point $(1, 0)$ and calculate the derivative $\frac{dy}{dt}$ at this point:

$$\frac{dy}{dt}(1, 0) = 2 \cdot 1 = 2 > 0.$$

As the derivative $\frac{dy}{dt} > 0$, then the point crosses the x -axis in the upward direction with increasing time t. We mark this on the phase plane. Then, using symmetry, we can easy specify the direction of movement for the other phase trajectories.

Example 2

Investigate the equilibrium positions of the linear autonomous system and draw its phase portrait:

$$\begin{cases} \frac{dx_1}{dt} = x_1 + 3x_2 \\ \frac{dx_2}{dt} = 2x_1 \end{cases}$$

Step 5

Draw the isoclines and separatrices on the phase plane and sketch the phase trajectories

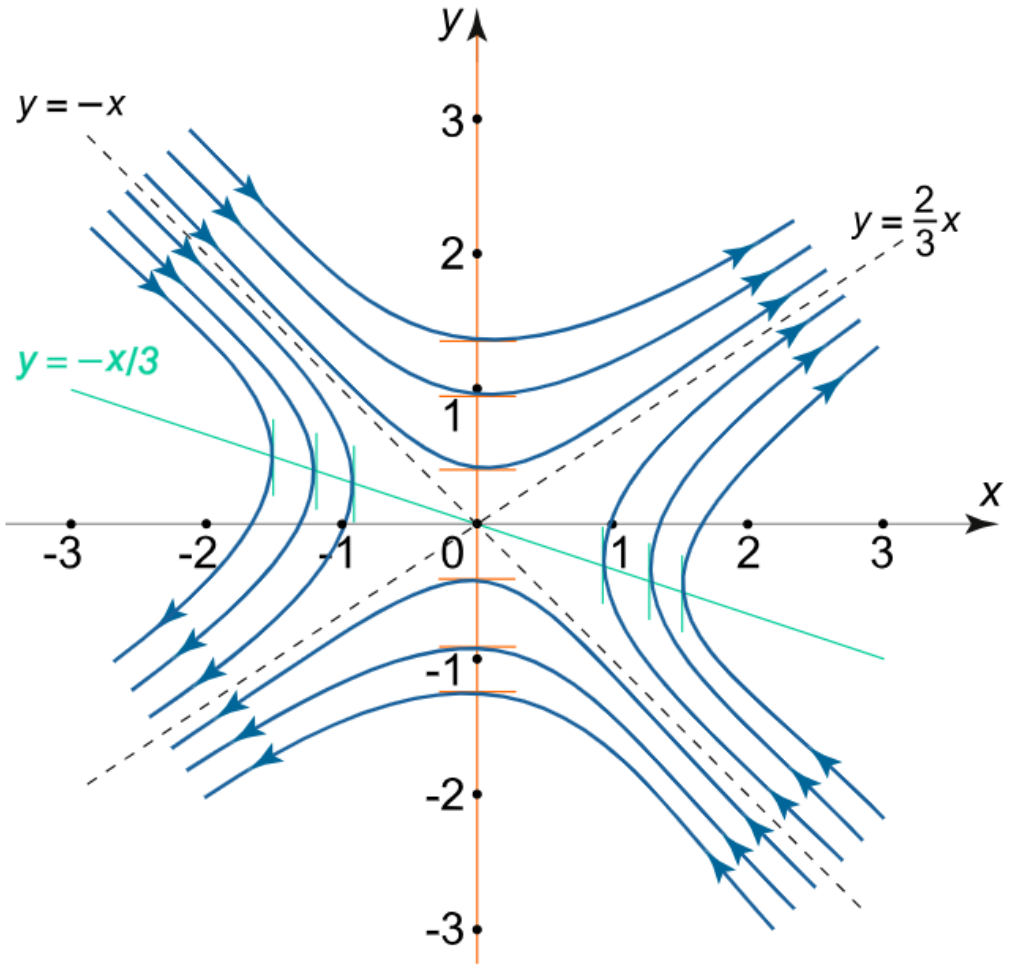
Determine the direction of motion along the phase trajectories.

Take, for example, the point $(1, 0)$ and calculate the derivative $\frac{dy}{dt}$ at this point:

$$\frac{dy}{dt}(1, 0) = 2 \cdot 1 = 2 > 0.$$

As the derivative $\frac{dy}{dt} > 0$, then the point crosses the x -axis in the upward direction with increasing time t. We mark this on the phase plane. Then, using symmetry, we can easy specify the direction of movement for the other phase trajectories.

Example 2



Example 3

Investigate the equilibrium positions of the dynamic system and sketch its phase portrait:

$$\begin{cases} \frac{dx_1}{dt} = 3x_1 - 4x_2 \\ \frac{dx_2}{dt} = 2x_1 - x_2 \end{cases}$$

Step 1

The determinant of this system is

$$A = \begin{bmatrix} 3 & -4 \\ 2 & -1 \end{bmatrix}, \quad \det A = \begin{vmatrix} 3 & -4 \\ 2 & -1 \end{vmatrix} = 5 \neq 0.$$

Hence, the system has the unique equilibrium point $(0, 0)$.

Example 3

Investigate the equilibrium positions of the dynamic system and sketch its phase portrait:

$$\begin{cases} \frac{dx_1}{dt} = 3x_1 - 4x_2 \\ \frac{dx_2}{dt} = 2x_1 - x_2 \end{cases}$$

Step 1

The determinant of this system is

$$A = \begin{bmatrix} 3 & -4 \\ 2 & -1 \end{bmatrix}, \quad \det A = \begin{vmatrix} 3 & -4 \\ 2 & -1 \end{vmatrix} = 5 \neq 0.$$

Hence, the system has the unique equilibrium point $(0, 0)$.

Example 3

Investigate the equilibrium positions of the linear autonomous system and draw its phase portrait:

$$\begin{cases} \frac{dx_1}{dt} = 3x_1 - 4x_2 \\ \frac{dx_2}{dt} = 2x_1 - x_2 \end{cases}$$

Step 2

We calculate the eigenvalues of A:

$$\begin{vmatrix} 3-\lambda & -4 \\ 2 & -1-\lambda \end{vmatrix} = 0, \ \Rightarrow (\lambda - 3)(\lambda + 1) + 8 = 0,$$

$$\Rightarrow \lambda^2 - 2\lambda + 5 = 0, \ D = -16, \ \Rightarrow \lambda_{1,2} = \frac{2 \pm 4i}{2} = 1 \pm 2i.$$

The eigenvalues λ_1, λ_2, are complex conjugate numbers with a positive real part. Therefore, the equilibrium position at the origin is an unstable focus.

Example 3

Investigate the equilibrium positions of the linear autonomous system and draw its phase portrait:

$$\begin{cases} \frac{dx_1}{dt} = 3x_1 - 4x_2 \\ \frac{dx_2}{dt} = 2x_1 - x_2 \end{cases}$$

Step 3

Define the equations of isoclines. The vertical isocline is described by the linear function:

$$\frac{dx}{dt} = 3x - 4y = 0, \ \Rightarrow y = \frac{3}{4}x.$$

The equation of the horizontal isocline is

$$\frac{dy}{dt} = 2x - y = 0, \ \Rightarrow y = 2x.$$

Example 3

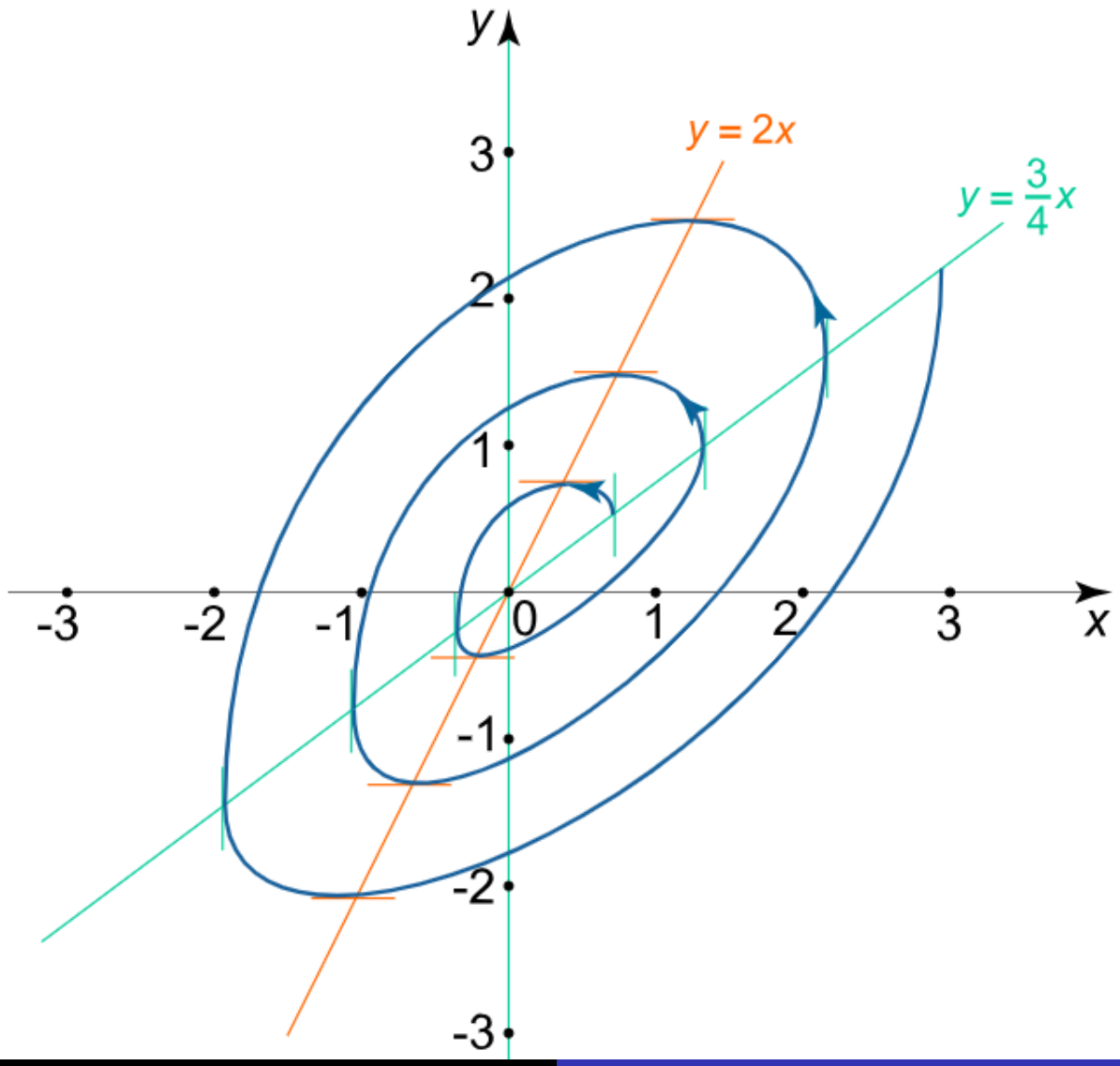
Investigate the equilibrium positions of the linear autonomous system and draw its phase portrait:

$$\begin{cases} \frac{dx_1}{dt} = 3x_1 - 4x_2 \\ \frac{dx_2}{dt} = 2x_1 - x_2 \end{cases}$$

Step 4

Find out the direction of twisting calculating the derivative $\frac{dy}{dt}$ at the point $(1, 0)$:

$$\frac{dy}{dt}(1, 0) = 2 \cdot 1 - 0 = 2 > 0.$$

Thus, the spirals twist counterclockwise.

Example 3